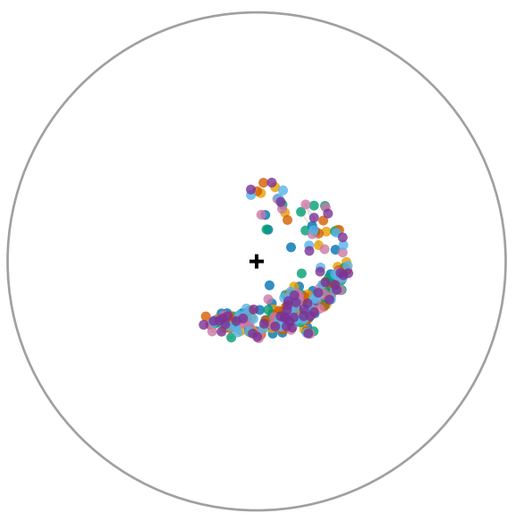
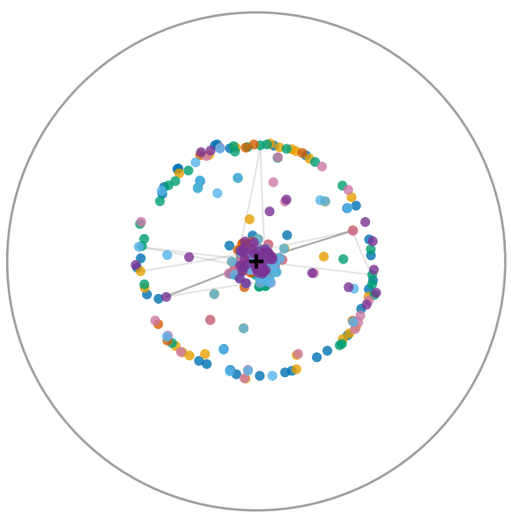
MSRVTT validation — hyperbolic + cone — 64D — training seed 50

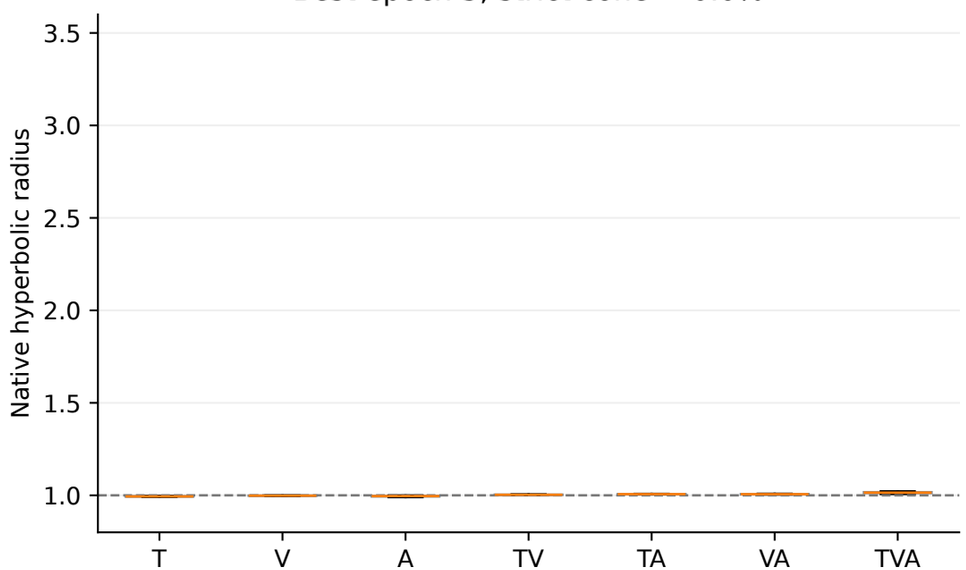
R0 — HoroPCA
distance ρ=0.44; kNN10=27.4%



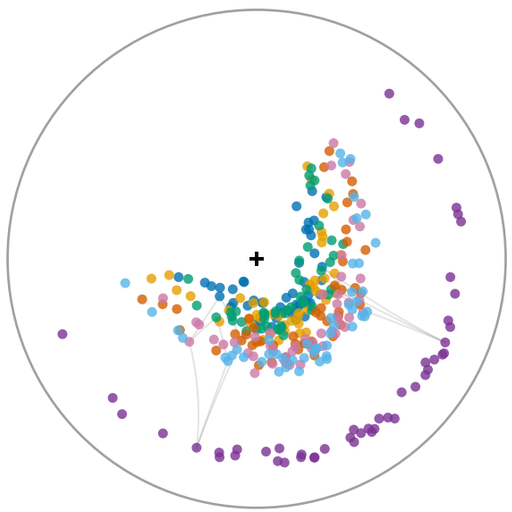
R0 — CO-SNE
distance ρ=-0.01; kNN10=15.4%



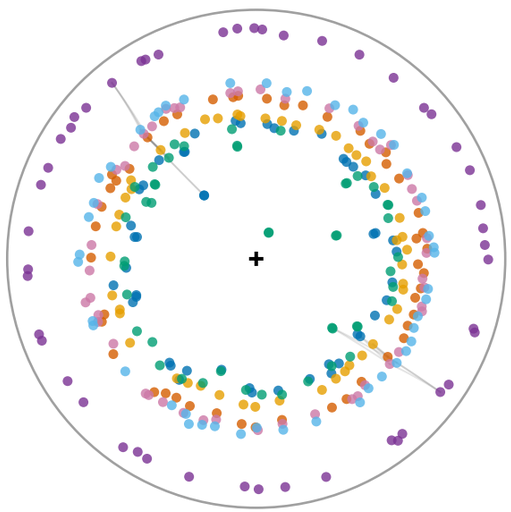
R0: 10-epoch baseline
Best epoch 5; strict cone = 0.0%



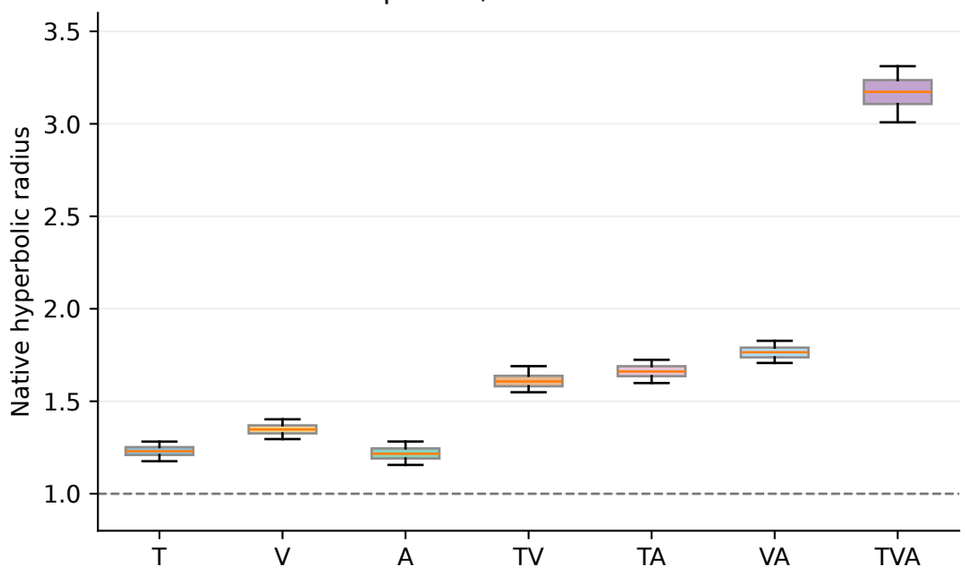
R3 — HoroPCA
distance ρ=0.73; kNN10=17.1%



R3 — CO-SNE
distance ρ=0.67; kNN10=36.0%



R3: head pretraining + fine-tuning
Best epoch 7; strict cone = 0.0%

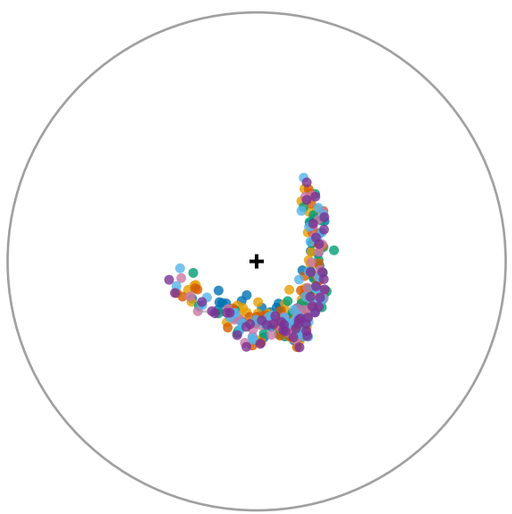


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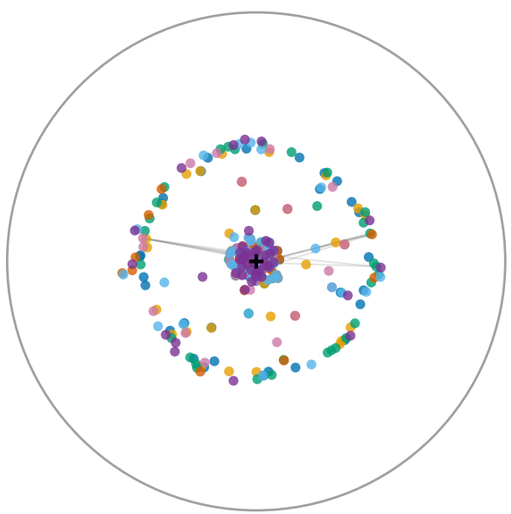
Same 48 validation videos × 7 subsets; projection seed 50. Radius plots: all 500 validation videos. Gray edges are prescribed relations, not confirmed containment.

MSRVTT validation — hyperbolic + cone — 64D — training seed 50

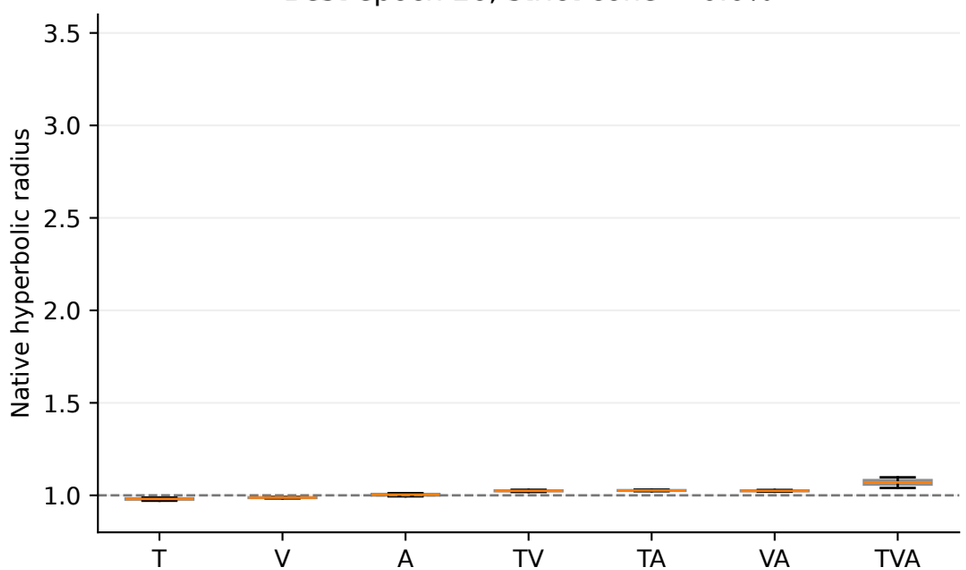
R1 — HoroPCA
distance ρ=0.37; kNN10=27.6%



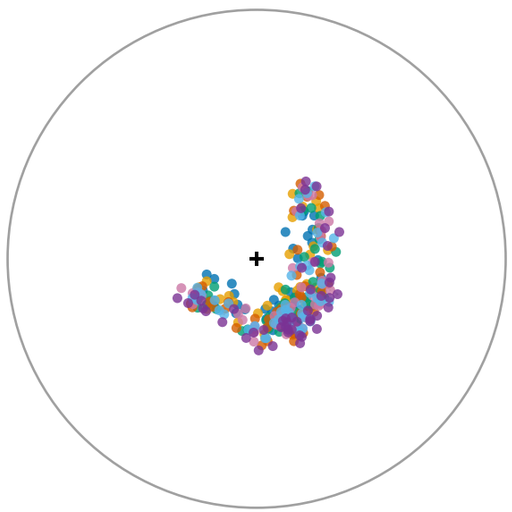
R1 — CO-SNE
distance ρ=0.00; kNN10=16.2%



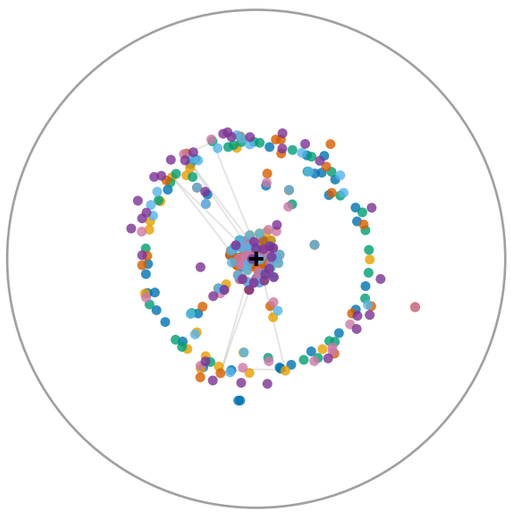
R1: 30-epoch fine-tuning
Best epoch 26; strict cone = 0.0%



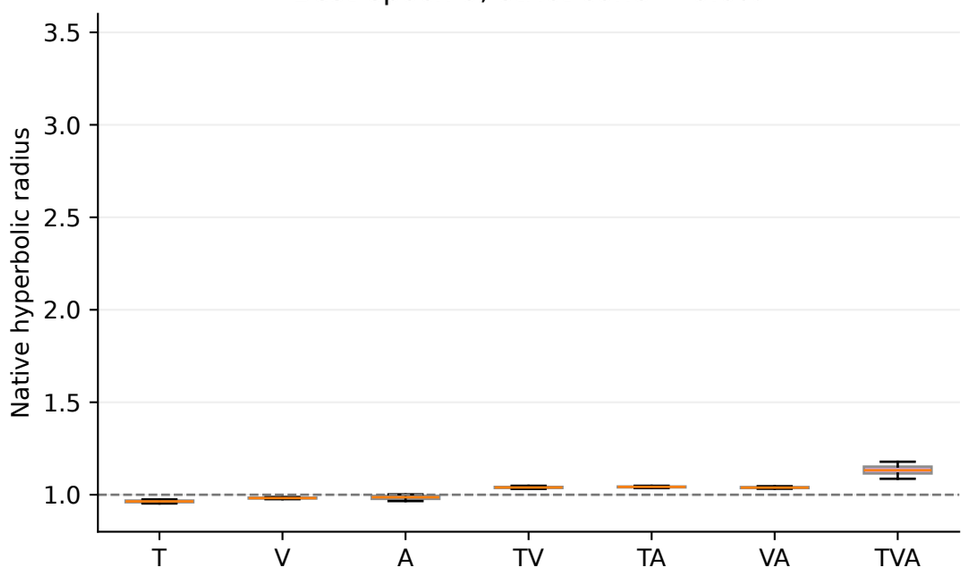
R2 — HoroPCA
distance ρ=0.42; kNN10=23.4%



R2 — CO-SNE
distance ρ=0.03; kNN10=16.0%



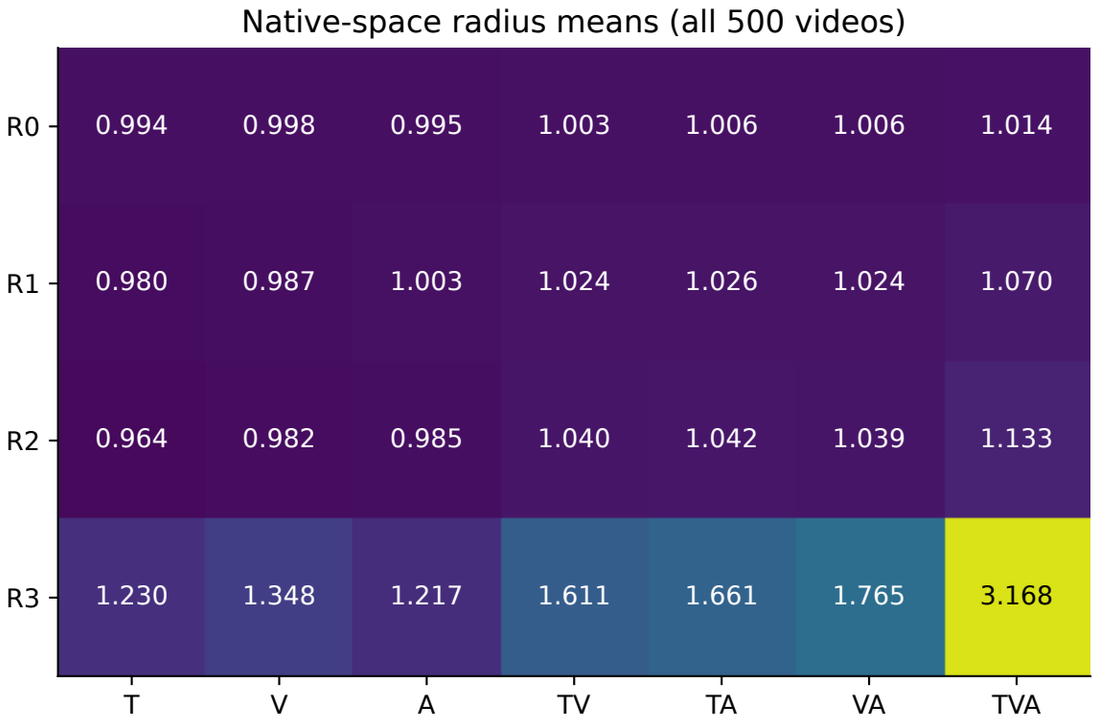
R2: higher head learning rate
Best epoch 9; strict cone = 0.0%

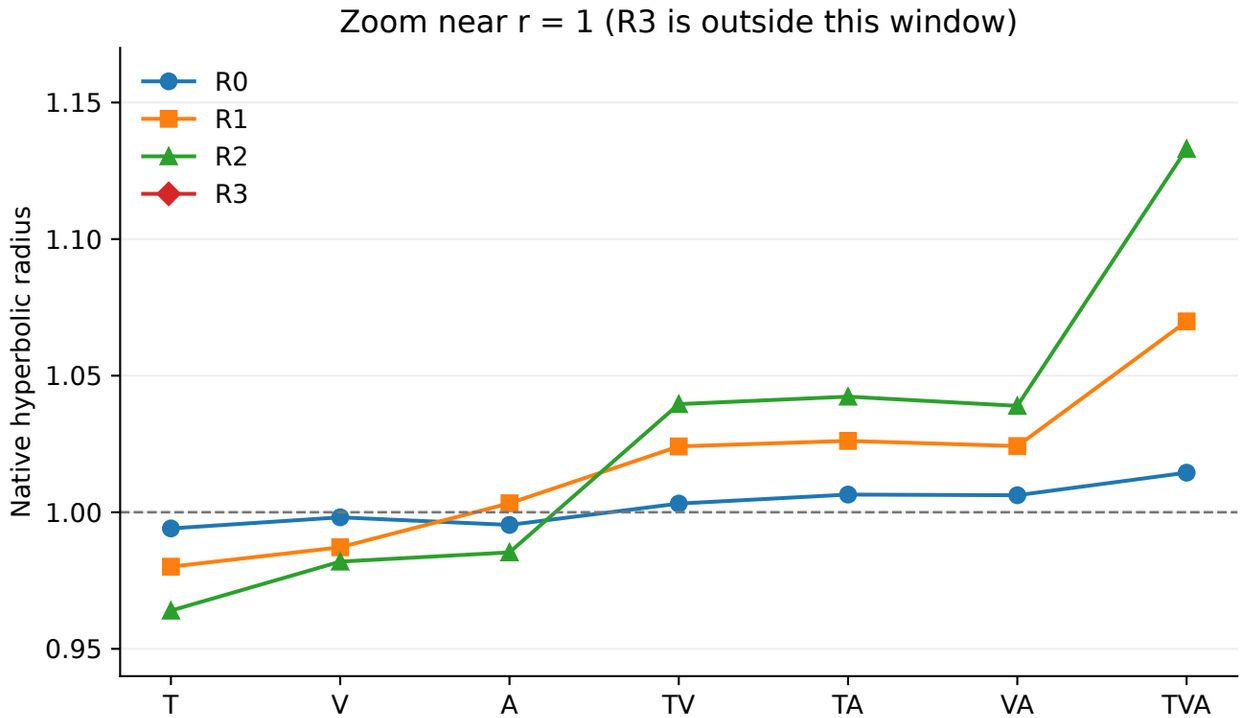


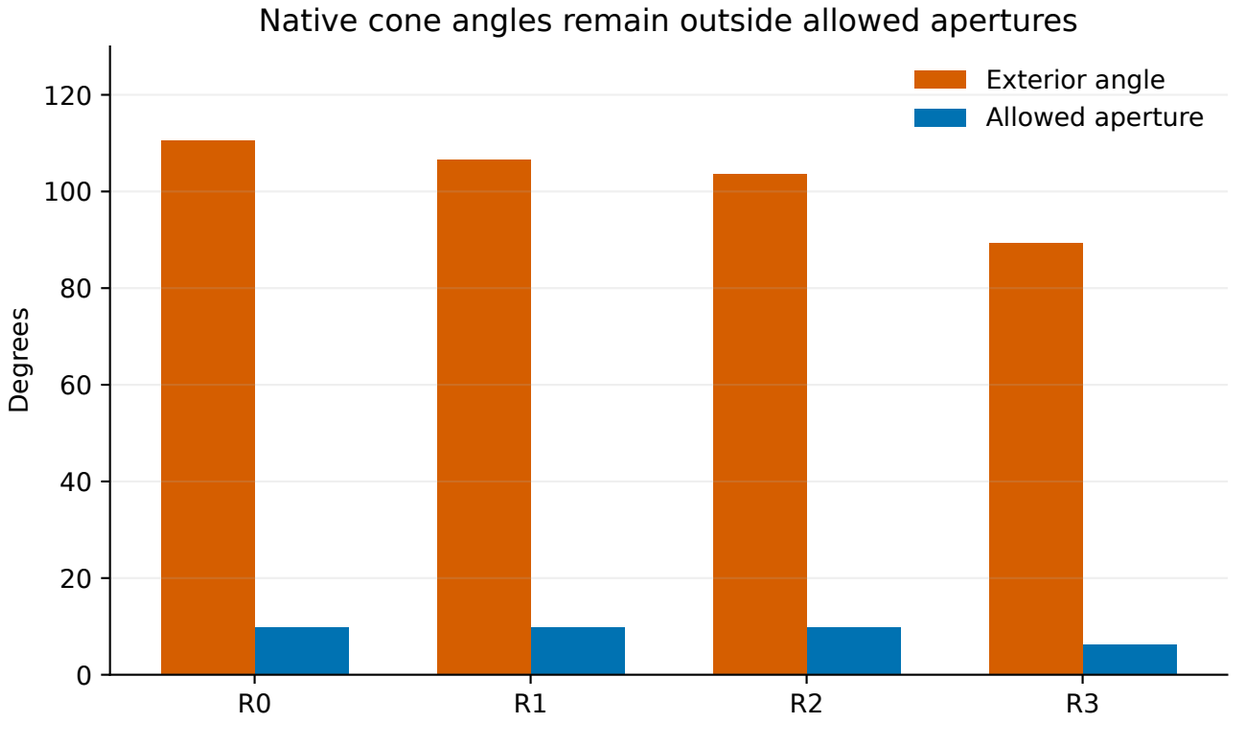
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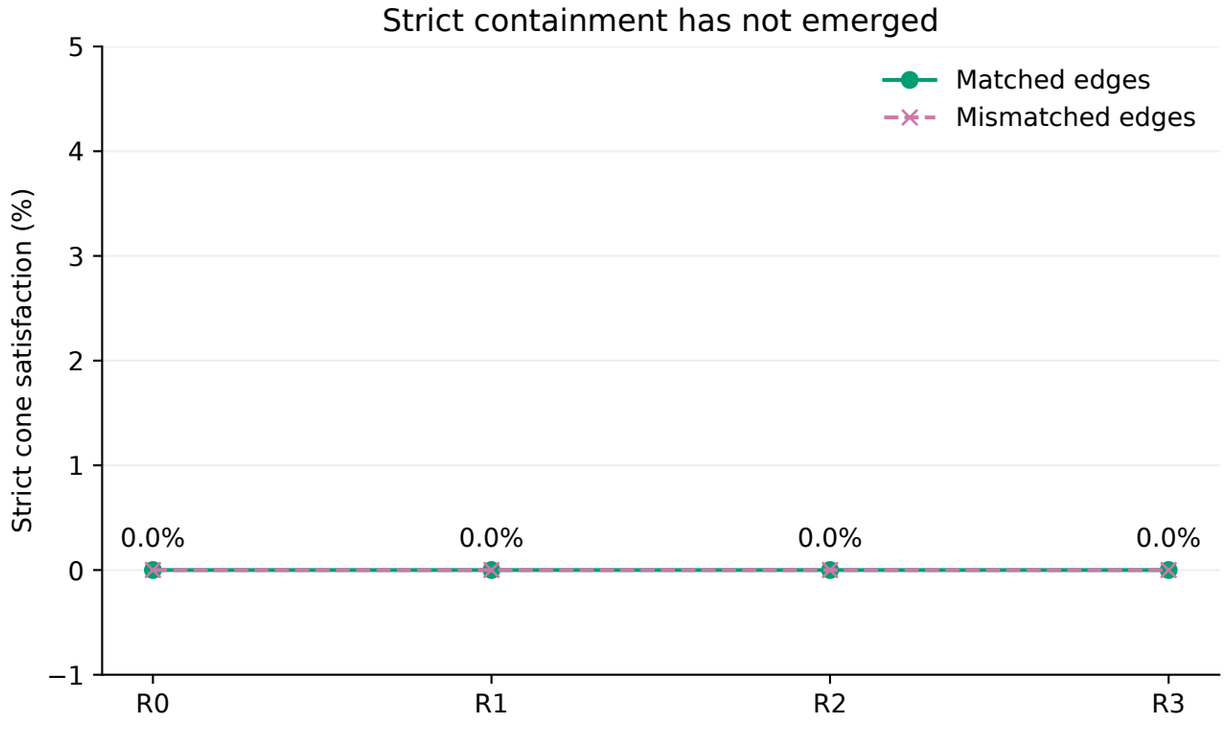
Same 48 validation videos × 7 subsets; projection seed 50. Radius plots: all 500 validation videos. Gray edges are prescribed relations, not confirmed containment.

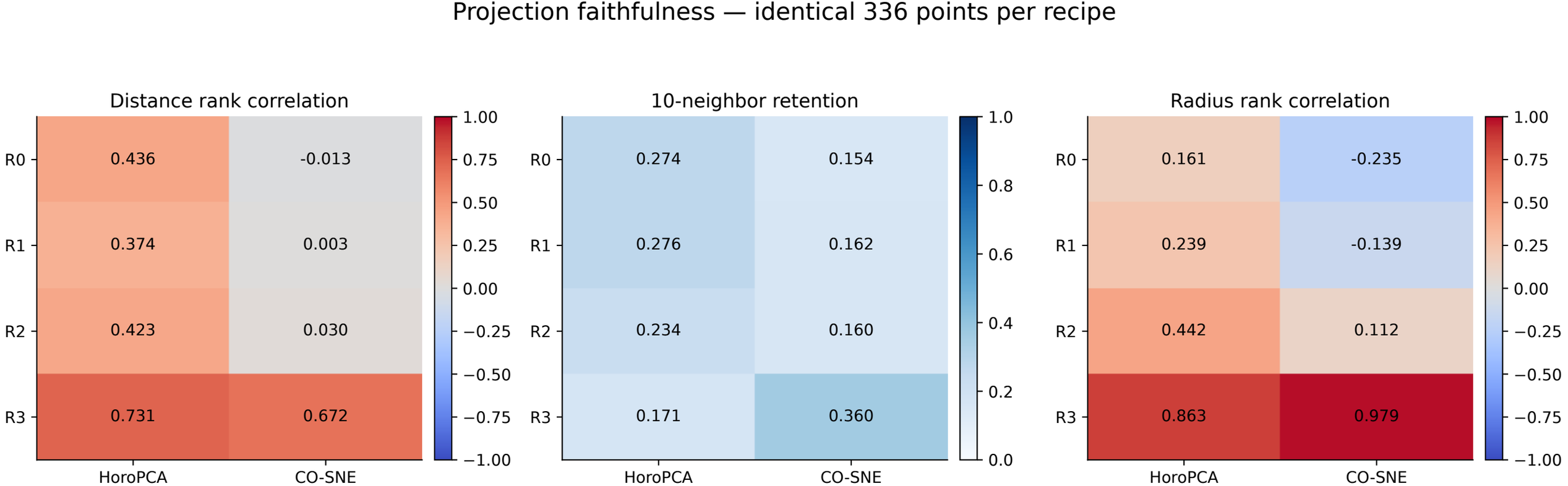
Native geometry — same validation cohort and retrieval-selected checkpoints










Projection faithfulness — identical 336 points per recipe
Distance rank correlation
R0
R1
R2
R3
0.436
-0.013
0.374
0.003
0.423
0.030
0.731
0.672
HoroPCA
CO-SNE
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
10-neighbor retention
0.274
0.154
0.276
0.162
0.234
0.160
0.171
0.360
1.0
0.8
0.6
0.4
0.2
0.0
Radius rank correlation
0.161
-0.235
0.239
-0.139
0.442
0.112
0.863
0.979
Low or negative radial correlation makes 2D distance from the center an unreliable indicator of learned radius.